

Real-time Air Quality Monitoring and Prediction using Machine Learning

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1. Abstract

Air pollution is a major environmental issue affecting public health and climate conditions. Monitoring air quality in real time helps governments and environmental agencies detect pollution spikes and take preventive actions.

This project develops a machine learning-based system to monitor and analyze air quality data. The dataset contains pollutant levels such as PM2.5, PM10, NO₂, SO₂, CO, and O₃ along with timestamps.

The system performs data preprocessing, exploratory data analysis, feature engineering, machine learning prediction, and pollution alert detection. A Random Forest Regressor model is used to predict the Air Quality Index (AQI).

The cleaned dataset is exported for Power BI visualization and also stored in MySQL database for real-time data management.

2. Introduction

Air pollution is one of the biggest environmental challenges worldwide. Pollutants like particulate matter (PM2.5, PM10), nitrogen dioxide (NO₂), sulphur dioxide (SO₂), carbon monoxide (CO), and ozone (O₃) significantly impact human health.

Traditional monitoring systems only display pollution levels but do not predict future air quality conditions.

This project builds a data analysis and machine learning pipeline that:

- Monitors air pollution levels
 - Predicts AQI using machine learning
 - Detects pollution alerts
 - Stores data in a database
 - Visualizes data using Power BI dashboards
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3. Goal of the Project

The main objectives of this project are:

- 1. Monitor air pollution levels using environmental data.**
 - 2. Perform data cleaning and preprocessing.**
 - 3. Analyze pollution patterns using data visualization.**
 - 4. Predict AQI using machine learning.**
 - 5. Detect high pollution alerts.**
 - 6. Store processed data in MySQL database.**
 - 7. Export clean dataset for Power BI dashboard visualization.**
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4. Data Overview

The dataset `real_time_air_quality_20k.csv` contains approximately 20,000 records of air pollution data.

Dataset Features

Column Description

timestamp Date and time of the reading

PM2.5 Fine particulate matter

PM10 Coarse particulate matter

NO2 Nitrogen dioxide level

SO2 Sulphur dioxide level

CO Carbon monoxide level

O3 Ozone level

AQI Air Quality Index

5. Data Preprocessing

Before building the machine learning model, the dataset undergoes several preprocessing steps.

5.1 Column Name Correction

Special characters in column names were removed to ensure compatibility with MySQL database.

Example:

PM2.5(Fine particulate matter) → PM25

PM10(Coarse particulate matter) → PM10

5.2 Timestamp Conversion

The timestamp column was converted into datetime format for time-based analysis.

```
df['timestamp'] = pd.to_datetime(df['timestamp'], dayfirst=True)
```

5.3 Duplicate Removal

Duplicate records were removed to maintain data quality.

```
df = df.drop_duplicates()
```

5.4 Missing Value Analysis

Missing values were checked using:

```
df.isnull().sum()
```

6. Feature Engineering

Additional time-based features were extracted from the timestamp column.

Feature Description

hour Hour of the day

day Day of the month

month Month of the year

Example:

```
df['hour'] = df['timestamp'].dt.hour
```

```
df['day'] = df['timestamp'].dt.day
```

```
df['month'] = df['timestamp'].dt.month
```

These features help the model capture daily and seasonal pollution patterns.

7. Exploratory Data Analysis (EDA)

AQI Distribution

A histogram was used to analyze the distribution of AQI values.

Insights:

- Shows frequency of different pollution levels.
- Helps identify pollution spikes.

Average AQI by Hour

A line plot shows how AQI changes throughout the day.

Insights:

- Pollution often peaks during traffic hours.
- Lower levels may occur late at night.

8. Machine Learning Model

A Random Forest Regressor model was used to predict AQI.

Input Features

- PM25
- PM10
- NO2

- SO2
- CO
- O3
- hour
- day
- month

Target Variable

AQI (Air Quality Index)

Data Splitting

The dataset was split into:

- 80% Training Data
- 20% Testing Data

```
train_test_split(test_size=0.2)
```

Feature Scaling

StandardScaler was used to normalize the features.

```
scaler = StandardScaler()
```

This improves model performance.

Model Training

The model used:

```
RandomForestRegressor(n_estimators=100)
```

Random Forest is chosen because:

- Handles nonlinear relationships
 - Works well with environmental data
 - Provides feature importance
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9. Model Evaluation

The model performance was evaluated using:

Mean Absolute Error (MAE)

Average absolute difference between predicted and actual AQI.

Root Mean Square Error (RMSE)

Measures prediction error magnitude.

R² Score

Indicates how well the model explains variance in AQI.

Example output:

Metric Meaning

MAE Average prediction error

RMSE Standard deviation of errors

R² Model accuracy

10. Pollution Alert Detection

A pollution alert system was implemented.

If AQI exceeds 150, the system flags High Pollution.

AQI_THRESHOLD = 150

Example output:

AQI Alert

90 Normal

180 High Pollution

This helps authorities detect pollution spikes instantly.

11. Feature Importance

Random Forest provides feature importance scores.

A bar chart shows which pollutants influence AQI most.

Typical important features:

- PM2.5
- PM10
- NO2
- O3

This helps identify major pollution contributors.

12. Export Dataset for Power BI

The cleaned dataset is exported as CSV.

```
df.to_csv("real_time_air_quality_clean.csv")
```

This dataset can be used to create interactive Power BI dashboards showing:

- AQI trends
 - Pollution alerts
 - Pollutant contribution
 - Hourly pollution patterns
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13. Python–MySQL Integration

The processed dataset is stored in a MySQL database.

Database Connection

```
engine = create_engine("mysql+pymysql://root:system@localhost/air_quality_db")
```

Data Insertion

```
df.to_sql(  
    name="air_quality",  
    con=engine,  
    if_exists="replace",  
    index=False
```

)

This enables:

- Real-time data storage
 - Database querying
 - Integration with analytics tools
-

14. Power BI Dashboard (Planned Visualizations)

Power BI dashboards can include:

1. AQI Trend Over Time

2. Pollution Alert Indicators

3. Average AQI by Hour

4. Pollutant Contribution Chart

5. AQI Distribution Histogram

These dashboards help policymakers make data-driven environmental decisions.

15. Conclusion

This project demonstrates how data analysis, machine learning, and database integration can be used to monitor air quality effectively.

The system:

- Processes real-time environmental data
- Predicts AQI using machine learning

- Detects pollution alerts
- Stores data in MySQL
- Supports visualization using Power BI

Such systems can help cities monitor pollution trends and take early preventive actions.

16. Future Improvements

Future enhancements could include:

- Real-time sensor integration
- Deep learning models for AQI prediction
- Live streaming dashboards
- SMS/email pollution alerts
- Integration with weather data